- Racing cars are built low and broad for stability.
- Table lamps and fans are designed with large heavy bases to make them stable.

Points to Remember

- ❖ The total length of a path taken by an object to reach one place from the another place is called distance.
- ❖ The shortest distance from the initial to the final position of an object.
- ❖ Acceleration is the rate of change in velocity. SI unit of acceleration is m/s^2.
- ❖ Velocity is the rate of change in displacement. SI unit of velocity is metre / second (m/s).
- ❖ The centre of gravity of an object is the point through which the entire weight of the object appears to act.
- ❖ Generally the centre of gravity of the geometrical shaped object lie on the geometric centre of the object.
- ❖ Stability is a measure of the body's ability to maintain its original position.
- ❖ The three types of stability are: stable equilibrium, unstable equilibrium, neutral equilibrium.

Evaluation

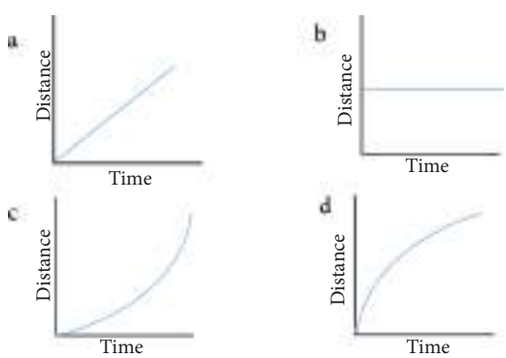
1LTED

I. Choose the best answer.

1. A particle is moving in a circular path of radius *r*. The displacement after half a circle would be

   a. Zero b. R c. 2 *r* d. *r* / 2

2. Which of the following figures represent uniform motion of a moving object correctly?

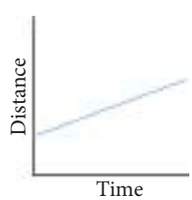


3. Suppose a boy is enjoying a ride on a merry go round which is moving with a constant speed of 10 m/s. It implies that the boy is

   a. at rest
   b. moving with no acceleration
   c. in accelerated motion
   d. moving with uniform velocity

4. From the given v-t graph it can be inferred that an object is

   a. in uniform motion b. at rest
   c. in non - uniform motion
   d. moving with uniform accelerations



5. How can we increase the stability of an object?

   a. Lowering the centre of gravity
   b. Raising the centre of gravity
   c. Increasing the height of the object
   d. Shortening the base of the object

II. Fill in the blanks.

1. The shortest distance between two places is _____.
2. The rate of change of velocity is_____.

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